

MAKE UP EXAMINATIONS – JUNE 2023

Program	: Science / Artificial Intelligence and Machine Learning	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: AD33 / AI33	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss how malloc(), calloc() and free() functions are used in dynamic memory allocation. CO1 (06)
 - Write a program in C to define a student with fields as USN, NAME, and ADDRESS. Declare Student variables in main () and demonstrate the accessing of structure members using dot operator. CO1 (06)
 - Write a C function to perform Fast Transpose of a Sparse Matrix. CO1 (08)
- Write a program to declare Strings and demonstrate strlen() function. CO1 (08)
 - Differentiate between struct and union with syntax. CO1 (06)
 - List the rules for declaring a pointer and operations on pointers. CO1 (06)

UNIT - II

- Write C functions to perform push() and pop() operations on a stack using array implementation. CO2 (06)
 - Illustrate various operations on multiple queues with neat diagrams. CO2 (06)
 - Apply infix to postfix conversion algorithm to convert following infix expressions to postfix expressions:
 - $((A-(B+C))*D)/(E+F)$
 - $A+(B-C*(D-E)/F)-G$
- Write an ADT for a queue and include necessary operations. CO2 (06)
 - Write a C function to compare contents of two linear queues using array implementation. CO2 (06)
 - Use the postfix evaluation algorithm to evaluate the following expression $723+-382/*-$ CO2 (08)

UNIT - III

- What is Doubly Linked List? Write an algorithm to insert and delete a node in Doubly Linked List. CO3 (06)
 - Represent the following two polynomials using singly linked lists and compute $C(x)=A(x)+B(x)$.

$$A(x)=5x^6+6x^4-4x+10$$

$$B(x)=4x^7+2x^3+7x^2-9$$
 - Write a C function to multiply every alternate node by 5 in a singly linked list and display the entire list after modification. CO3 (06)

6. a) Write a function INVERT that inverts a singly linked list and returns a pointer to this new list. Write the following assumptions made. CO3 (06)
b) Write a function to delete every alternate node in a singly linked list. CO3 (08)
c) Write C functions for push() and pop() operations for a linked stack. CO3 (06)

UNIT- IV

7. a) Write a C function to perform Iterative Inorder traversal of a binary tree using stack. CO4 (06)
b) Suppose that we have the following key values: 7, 16, 49, 82, 5, 31, 6, 2, and 44. Write out the MAX heap after each value is inserted into the heap CO4 (08)
c) Write a non-recursive C function to check whether two binary trees are equal or not. CO4 (06)
8. a) Define the following with an example: CO4 (06)
i) Binary tree
ii) Binary search tree
iii) Complete binary tree.
b) Write a C function to perform Inorder traversal of a threaded Binary tree. CO4 (06)
c) Construct binary search tree for following values stepwise 10, 2, 14, 9, 3, 18, 1, 16, 10, 4, 11, 3. Perform inorder, preorder and postorder traversals on a binary search tree. CO4 (08)

UNIT - V

9. a) Write a C function to create a graph using adjacency list representation. CO5 (08)
b) Explain RR rotation of AVL tree with a neat diagram. CO5 (06)
c) Write a C function to implement Breadth First search of a Graph. CO5 (06)
10. a) Consider the following list of key values: CO5 (07)
40, 50, 70, 10, 20, 30, 60, 80, 90
Write the AVL tree after each value is inserted into the tree.
b) Write a C function to create a graph using an adjacency matrix representation. CO5 (08)
c) Write ADT for graph and include necessary operations. CO5 (05)
